

Positive Square Energy of Theta Graphs with One Tree Attachment

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Abstract

We prove that if G is a connected graph whose 2-core is a simple theta graph $\Theta(a, b, c)$ and one arbitrary rooted tree is attached to a single core vertex through one bridge edge, then $s^+(G) > |V(G)|$. This extends the bare theta theorem of our companion paper to the weighted-core setting of Conjecture 1.2 of Akbari et al. (arXiv:2506.07264). The proof combines an exact root-congruence PSD witness, a local four-fifths reduction via polynomial majorants, and a complete phase-area classification of the signed square trace by odd-girth residue class.

1 Introduction

For a graph G with adjacency eigenvalues λ_i , let $s^+(G) = \sum_{\lambda_i > 0} \lambda_i^2$. The companion paper proves that every simple theta graph G satisfies $s^+(G) > |V(G)|$. Here we extend this to one arbitrary rooted-tree attachment.

Theorem 1. *Let H be a simple theta graph and let $v \in V(H)$. Let T be a rooted tree with root r of degree one in $T - r$, and let G be obtained by identifying r with v and adding one bridge edge from r to a new vertex. Then*

$$s^+(G) > |V(G)|.$$

More precisely, for every root v of H ,

$$s^+(A(H) - \tfrac{1}{2}E_{vv}) > |V(H)|.$$

The weighted 2-core reduction of Akbari et al. reduces the attachment problem to the endpoint $d = 1/2$ because $s^+(A - dE_{vv})$ is nonincreasing in d and the rooted-tree penalty $d = (A^-(T))_{vv} \leq 1/2$.

2 The root-congruence witness

Let A be the adjacency matrix of H , $P = A_+$, and fix a root v . Put $S = \text{tr}(P^2) = s^+(H)$, $a = P_{vv}$, $q = (P^2)_{vv}$. For $0 \leq k \leq 1$, set $C_k = I - (1 - k)E_{vv}$ and $X_k = C_k P C_k \succeq 0$.

Lemma 2.

$$s^+(A - \tfrac{1}{2}E_{vv}) \geq S - \frac{2(1-k)^2q}{1} - k^2a - (1-k^2)^2a^2.$$

At $k = 6/7$, this gives

$$s^+(A - \tfrac{1}{2}E_{vv}) \geq S - L, \quad L = \frac{2q}{49} + \frac{36a}{49} + \frac{169a^2}{2401}.$$

The proof is direct trace expansion using $AP = P^2$ and $A_{vv} = 0$. The exact certificate is in `root_congruence_certificate.py`.

3 Local four-fifths reduction

Lemma 3. *For every rooted simple theta, $L < 4/5$. More precisely, $4/5 - L \geq 31684811/1355316480 > 0$.*

Proof. The quadratic $V_t(x) = t/4 + x/2 + x^2/(4t) \geq x_+$ for all x , giving $a \leq t/4 + \deg(v)/(4t)$. The cubic $U(x) = 1/5 + x/2 + x^2/2 + x^3/9 \geq x_+^2$ on $[-3, 3]$, giving $q \leq 1/5 + \deg(v)/2 + \tau/9$, where $\tau = (A^3)_{vv} = 2T_v$. The five possible $(\deg(v), \tau)$ states are $(3, 0), (3, 2), (3, 4), (2, 0), (2, 2)$. Substituting the sharp bounds gives strict rational margins below $4/5$ in all five cases. $\square$

The exact certificate is in `local_four_fifths_certificate.py`. Together, Lemmas 2–3 prove:

Corollary 4. *If $s^+(H) - |V(H)| \geq 4/5$, then $s^+(A(H) - E_{vv}/2) > |V(H)|$ for every root v .*

4 The bare four-fifths theorem

Theorem 5. *Every simple theta graph H satisfies $s^+(H) - |V(H)| \geq 4/5$, except exactly $\Theta(2, 2, 3)$, $\Theta(2, 3, 3)$, and $\Theta(1, 4, 4)$.*

The proof uses the imaginary-axis phase representation. For $z = e^{-2u} \in (0, 1)$ and $x = 2i \sinh u$, the theta characteristic polynomial factors through a phase carrier

$$H(z) = R(z) + i\sqrt{z} S(z),$$

where $R \geq 0$ (in fact $R > 0$ for nonbipartite thetas) and

$$\text{tr}(A|A|) = -\frac{2}{\pi} \int_0^1 (z^{-2} - 1) \text{Arg } H(z) dz.$$

Since $s^+ - n = 1 + \text{tr}(A|A|)/2$, the target $4/5$ is equivalent to

$$\int_0^1 (z^{-2} - 1) \alpha(z) dz \leq \frac{\pi}{5}.$$

Phase sign classification

The shortest odd and even path lengths o, e determine $\text{sign}(PQ) = (-1)^{(o+e-1)/2}$. If the shortest odd cycle $g = o + e \equiv 3 \pmod{4}$, then $\text{tr}(A|A|) > 0$ and $s^+ - n > 1$. Bipartite thetas have $s^+ - n = 1$. Thus only $g \equiv 1 \pmod{4}$ requires quantitative control.

The $\tilde{C}_5$ class

When $g = 5$, the only residue pairs are $(1, 4)$ and $(2, 3)$. Exact mod-four phase monotonicity (`theta_phase_monotonicity_certificate.py`), safe common limits ($I_\infty < 8/15 < \pi/5$), and exact Sturm gates (`theta_short_base_gates.py`) settle both channels outside $\Theta(2, 2, 3)$, $\Theta(2, 3, 3)$, $\Theta(1, 4, 4)$.

Odd girth at least nine

When $g \geq 9$, a 16-case parameterized certificate (`theta_g9_signed_certificate.py`) proves $K = 4z^4 R - S \geq 0$ on $[0, 3/4] \times [0, 1]^3$ by exact tensor Bernstein coefficients, giving $\alpha(z) < 4z^{9/2}$ on $[0, 3/4]$. On $[3/4, 1]$, $\alpha \leq \pi/2$ and the tail weight has mass $1/12$. Thus $I < 3051\sqrt{3}/19712 + \pi/24 < \pi/5$.

5 Exceptional weighted endpoints

For the three bare exceptions, direct Sturm isolation of $\chi(A - E_{vv}/2)$ at every root orbit proves $s^+(A - E_{vv}/2) > |V(H)|$ with exact rational margins. The complete certificate covering all 9 root orbits is in `theta_short_base_gates.py`.

6 Conclusion

Theorem 5 plus Corollary 4 prove Theorem 1 for every nonexceptional theta. The three exceptions are covered by their direct weighted certificates. This completes the one-tree weighted-theta extension.

References

- [1] S. Akbari, H. Kumar, B. Mohar, S. Pragada, and S. Zhang, *Refinement of a conjecture on positive square energy of graphs*, arXiv:2506.07264v1, 2025.